Multiple linear regression, variable selection and regularisation [MCQ] (Version : 1)

TEST

Correct Answer

Answered in 2.18 Minutes

Question 1/10

A data scientist applies Lasso regularization (L1) to a multiple linear regression model in an attempt to improve model performance. After tuning the regularization strength, they observe that several coefficients of predictor variables have been reduced to zero. What does this outcome indicate about the affected predictor variables?

- [ ] The regularization strength was too low, and increasing it would bring the coefficients back to non-zero values.
- [ ] These variables are highly correlated with the dependent variable and should be kept in the model regardless of their coefficients.
- [ ] Lasso regularization has incorrectly identified these variables as insignificant due to a scaling issue in the dataset.
- [x] These variables are likely not contributing significantly to the model and can be considered for removal to improve model simplicity and interpretability.

Question 2/10

What does RSS stand for?

- [ ] Residual Squared State

- [ ] Regression Sum of Squares
- [ ] Regression Squared State
- [x] Residual Sum of Squares

Question 3/10

Compare the three models below:

Model 1: train MSE = 0.423, test MSE = 0.978 Model 2: train MSE = 0.572, test MSE = 0.644 Model 3: train MSE = 0.218, test MSE = 1.103

Based on this information, which of these models generalises the best to unseen data?

- [ ] Model 3
- [x] Model 2
- [ ] There is no indication of one being better than the others
- [ ] Model 1

Question 4/10

Compare the three models below:

Model 1: train MSE = 0.323, test MSE = 0.812 Model 2: train MSE = 0.682, test MSE = 0.694 Model 3: train MSE = 0.257, test MSE = 1.112

Based on this information, which of these models generalises the worst to unseen data?

- [x] Model 3

- [ ] Model 2
- [ ] There is no indication of one being worse than the others
- [ ] Model 1

Question 5/10

When adding more variables to a linear model, what is true about the R-squared value?

- [x] Adding more variables will always increase R-squared.
- [ ] R-squared cannot be calculated for models that use multiple predictor variables.
- [ ] R-squared is not affected by adding additional variables.
- [ ] Adding more variables will always decrease R-squared.

Question 6/10

Adding more variables always leads to a lower test MSE.

- [ ] True
- [x] False

Question 7/10

What does RMSE stand for?

- [ ] Random Mean Sampling Error
- [ ] Really Mean Statistical Error
- [x] Root Mean Squared Error
- [ ] Root Median Sum of Errors

Question 8/10

The R-squared measure is said to take on a 'proportion' of some attribute associated with the model. What is that proportion?

- [ ] Proportion of observations used for training.
- [ ] Proportion of predictor variables contributing to output.
- [ ] Proportion of outputs correctly predicted.
- [x] Proportion of variance explained.

Question 9/10

When our predictor variables have ranges and units that are quite different, it is pertinent to scale them before using them in a regression. Which of the following statements regarding scaling is FALSE?

- [ ] Normalisation is the process of squeezing a range of values to into the range [0,1].
- [ ] Normalisation is a scaling process which is

- [ ] inherently sensitive to outliers in the data.
- [ ] Standardisation centres and scales a set of values such that they all have a mean of 0 and standard deviation of 1.
- [x] Standardisation is the process of squeezing a range of values to into the range [0,1].

Question 10/10

Ridge and LASSO regression are known as regularisation techniques which we can use to improve a linear regression model. Which statement regarding these techniques is FALSE?

- [ ] LASSO regression attempts to minimise the RSS as well as the L1-norm.
- [ ] Both methods shrink the magnitude of variable coefficients, but LASSO can shrink the values all the way to zero.
- [x] Both ridge and LASSO regression are able to shrink coefficients all the way to zero.
- [ ] Ridge regression attempts to minimise the RSS as well as the L2-norm.